

Close

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
PYTHON FOR DATA SCIENCE Practical
Practical

Roll No.: S017
Name: KHAN IQRA
Class: SYBsc IT
Subject: PYTHON FOR DATA SCIENCE
Date of Assignment: 23/09/2026
Date/Time of Submission: 23/09/2026

Laxmi Charitable Trust's Sheth L.U.J College of Arts & Sir M.V.
College of Science and Commerce Department of Information
Technology (B.Sc.I.T Semester III) PYTHON FOR DATA SCIENCE
Practical Practical – VIII

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PART – B Practical Questions

BeautifulSoup

1. Extract Headings

Write a Python program using BeautifulSoup to extract and print all <h2> headings from a webpage. **bold text**

```
!pip install requests beautifulsoup4
```

```
Requirement already satisfied: requests in /usr/local/lib/python3.13/dist-packages (2.32.4)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.13/dist-packages (4.13.5)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/dist-packages (from requests) (3.19)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests) (2026.7.22)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.13/dist-packages (from beautifulsoup4) (2.9.2)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.13/dist-packages (from beautifulsoup4) (4.13.5)
```

```
import requests
from bs4 import BeautifulSoup

url = "https://www.geeksforgeeks.org/"
soup = BeautifulSoup(requests.get(url).text, "html.parser")

for h in soup.find_all("h2"):
    print(h.text.strip())
```

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2. Extract Links

Write a Python program using BeautifulSoup to extract all <a> tags and print their link text and URLs. **bold text**

```
!pip install requests beautifulsoup4
```

```
Requirement already satisfied: requests in /usr/local/lib/python3.13/dist-packages (2.32.4)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.13/dist-packages (4.13.5)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests) (3.4.9)
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Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.13/dist-packages (from beautifulsoup4) (4.13.5)
```

```
import requests
from bs4 import BeautifulSoup

url = "https://www.google.com"
soup = BeautifulSoup(requests.get(url).text, "html.parser")

for a in soup.find_all("a"):
    print(a.text.strip(), "->", a.get("href"))
```

Gmail -> <https://mail.google.com/mail/&ogbl>
Images -> <https://www.google.com/imghp?hl=en&ogbl>
-> <https://www.google.com/intl/en/about/products>
Sign in -> <https://accounts.google.com/ServiceLogin?hl=en&passive=true&continue=https://www.google.com/&ec=GAZAmgQ>

```
Advanced search -> /advanced_search?hl=en&authuser=0
Advertising -> /intl/en/ads/
Business Solutions -> /services/
About Google -> /intl/en/about.html
Privacy -> /intl/en/policies/privacy/
Terms -> /intl/en/policies/terms/
```

Seaborn

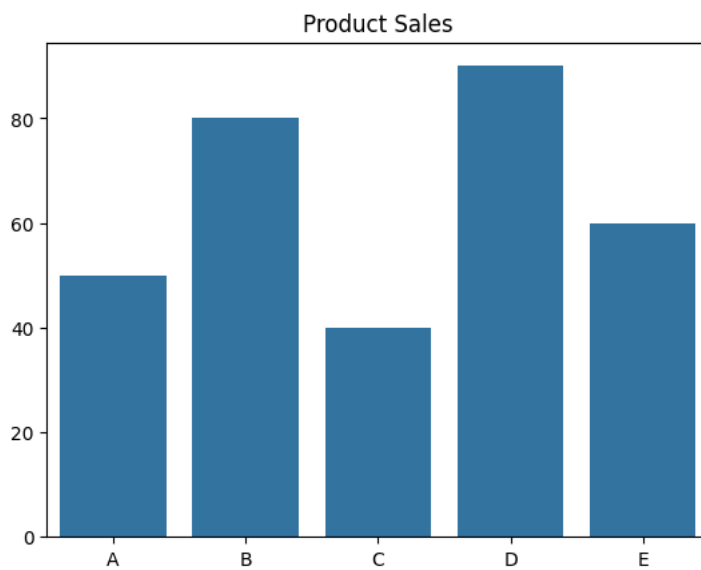
3. Bar Chart

Create a Seaborn bar chart showing the sales of 5 different products. **bold text**

```
import seaborn as sns
import matplotlib.pyplot as plt

products = ["A", "B", "C", "D", "E"]
sales = [50, 80, 40, 90, 60]

sns.barplot(x=products, y=sales)
plt.title("Product Sales")
plt.show()
```



4. Scatter Plot

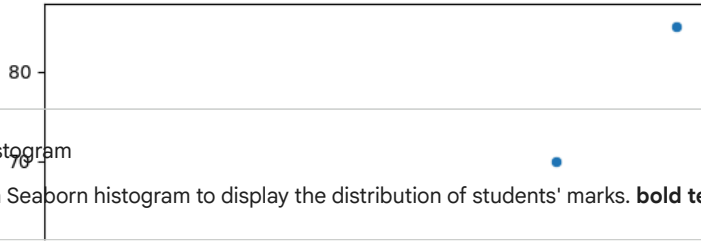
Create a Seaborn scatter plot showing the relationship between students' study hours and their marks. **bold text**

```
import seaborn as sns
import matplotlib.pyplot as plt

hours = [1, 2, 3, 4, 5, 6]
marks = [35, 45, 50, 60, 70, 85]

sns.scatterplot(x=hours, y=marks)
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Study Hours vs Marks")
plt.show()
```

Study Hours vs Marks



5. Histogram

Create a Seaborn histogram to display the distribution of students' marks. **bold text**

```
import seaborn as sns
import matplotlib.pyplot as plt

marks = [35, 45, 50, 55, 60, 65, 70, 72, 75, 80, 85, 90, 95]

sns.histplot(marks, bins=5, kde=True)
plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.title("Distribution of Marks")
plt.show()
```

